

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. CHB-501: Analytical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Differentiate between metaperiodate (IO_4^-) and paraperiodate (H_5IO_6) structures.
- (b) Define absolute error and relative error.
- (c) What are indicators in complexometric titrations?
- (d) Discuss the function of starch as an indicator in iodometry.
- (e) What is the role of masking agents in analytical chemistry?

Question 2.

- (a) Discuss the classification and minimization of errors in chemical analysis. [6]
- (b) What do you mean by significant figures? Apply computation rules to the following calculations so that significant figures are retained in the final results: [8]
 - (i) $3.4 + 0.020 + 7.31 = ?$
 - (ii) $(2.4 \times 4.02)/0.965 = ?$
 - (iii) $\log(4.000 \times 10^{-5}) = ?$
 - (iv) $\text{antilog}(12.5) = ?$

Question 3.

- (a) Name the titrations which you recommend for the quantitative evaluation of organic functional groups in the following reactions, and write chemical equations: [8]
 - (i) Acetylation of primary alcohols using acetic anhydride
 - (ii) Estimation of aldehydes using hydroxylamine hydrochloride
- (b) Discuss the theoretical and practical aspects of Karl Fischer reagent for the estimation of water content. [6]

Question 4.

- (a) Discuss the structural features of EDTA as a chelating agent in complexometric titrations. [6]
- (b) Explain with specific examples: [8]
- (i) EDTA displacement titration
 - (ii) EDTA back-titration
 - (iii) Metallochromic indicator transition mechanisms
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Question 5. Write explanatory notes on any two of the following: [2 × 7 = 14]

- (a) Global warming and its mitigation strategies
- (b) Ozone depletion mechanism in the stratosphere
- (c) Bhopal Gas Tragedy and industrial disaster management